

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460246.8312 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.052
Transit depth (Rp/Rs)^2	2.36 +/- 1.58 [%]
Orbital Inclination (inc)	85.21 +/- 1.61
Airmass coefficient 1 (a1)	0.7 +/- 0.5
Airmass coefficient 2 (a2)	0.19 +/- 0.33
Scatter in the residuals of the lightcurve fit is	72.94 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	3.0
Transit Duration (day)	0.04 +/- 0.043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.052 vs 0.1418 (deviation 0.14σ)
Duration fit vs published	0.04 d vs 0.1196 d (deviation 1.1σ)
Mid-time O–C	1.2 min
Depth SNR	1.8
Residual scatter	72.94 %

Light curve

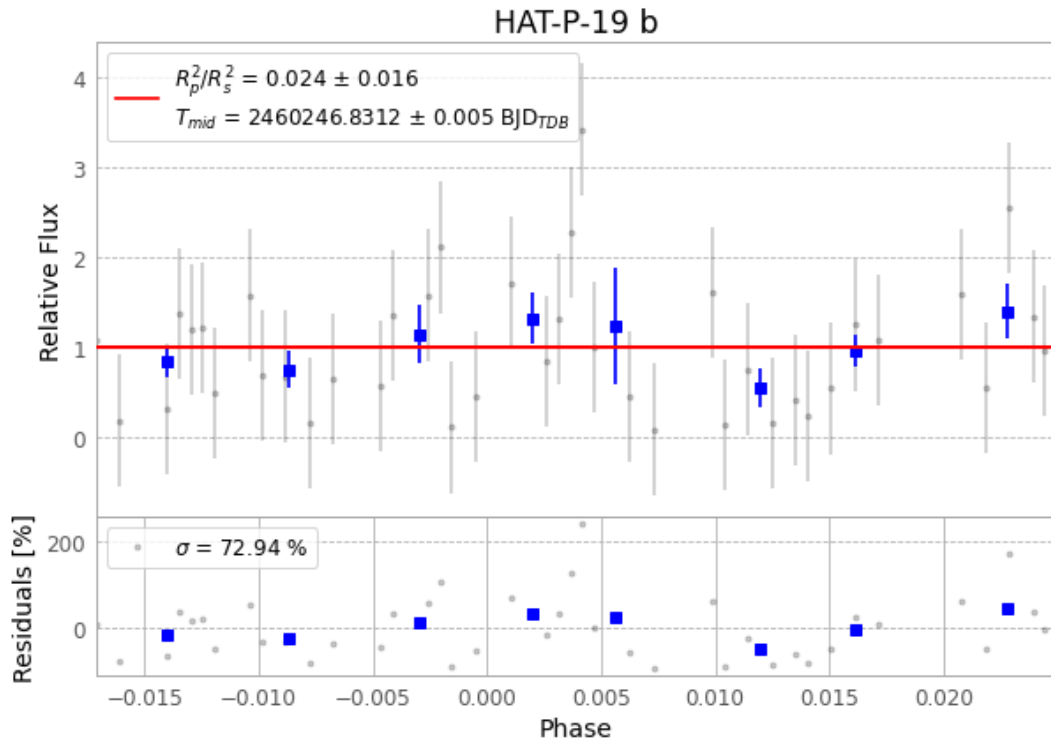


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-10-28.png (SHA-256 50866a56149e572e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [208, 430], 1 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 07db43436cc3cede...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), HATP-19231029060915.FITS → HATP-19231029101516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-231029.log (9e16df9749f1...), FinalLightCurve_HAT-P-19 b_2023-10-28.png (50866a56149e...), FinalLightCurve_HAT-P-19 b_2023-10-28.csv (cd0d5e82a883...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 05:27Z.